Clinical Biochemistry E-Book

67 Lipoprotein metabolism

Lipoproteins solve the problem of transporting fats around the body in the aqueous environment of the plasma. A lipoprotein is a complex spherical structure that has a hydrophobic core wrapped in a hydrophilic coating (Fig. 67.1). The core contains triglyceride and cholesteryl esters, and the surface contains phospholipid, free cholesterol and proteins – the apolipoproteins (Table 67.1). Cholesterol is an essential component of all cell membranes and is a precursor for steroid hormone and bile acid biosynthesis. Triglyceride is central to the storage and transport of energy within the body.

Nomenclature

Several different classes of lipoproteins exist the structure and function of which are closely related. Apart from the largest species, the chylomicron, these are named according to their density, as they are most commonly isolated by ultracentrifugation. The four main lipoproteins and their functions are shown in Table 67.2.

Metabolism

Lipoprotein metabolism (Fig. 67.2) can be thought of as two cycles, one exogenous and one endogenous, both centred on the liver. These cycles are interconnected.

Two key enzyme systems are involved in lipoprotein metabolism:

- *Lipoprotein lipase* (LPL) releases free fatty acids and glycerol from chylomicrons and very-low-density lipoprotein (VLDL) into the blood and thence into the tissues.
- *Lecithin–cholesterol acyltransferase* (LCAT) forms cholesteryl esters from free cholesterol and fatty acids.

The exogenous lipid cycle

Dietary lipid is absorbed in the small intestine and incorporated into chylomicrons that are secreted into the lymphatics and reach the bloodstream via the thoracic duct. In the circulation, triglyceride is gradually removed from these lipoproteins by the action of LPL. This enzyme is present in the capillaries of a number of tissues, predominantly adipose tissue and skeletal muscle. As it loses triglyceride, the chylomicron becomes smaller and deflated, with folds of redundant surface material. These remnants are removed by the liver. The cholesterol may be utilised by the liver to form cell membrane components or bile acids or may be excreted in the bile. The liver provides the only route by which cholesterol leaves the body in significant amounts.

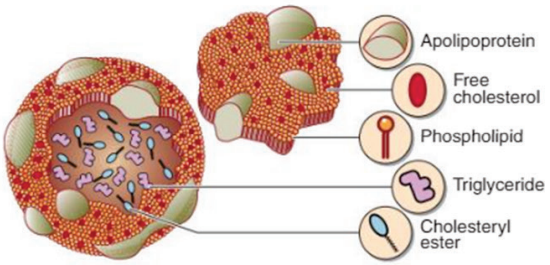


Fig. 67.1 Structure of a lipoprotein.

Table 67.1 Properties of some human apolipoproteins

Apolipoprotein	Molecular weight	Site of synthesis	Function
A-I	28000	Intestine, liver	Activates LCAT
A-II	17000	Intestine, liver	–
B_{100}	549000	Liver	Triglyceride and cholesterol transport. Binds to LDL receptor
B_{48}	264000	Intestine	Triglyceride transport
C-I	6600	Liver	Activates LCAT
C-II	8850	Liver	Activates LPL
C-III	8800	Liver	? Inhibits LPL
E	34000	Liver, intestine, macrophage	Binds to LDL receptor and probably also to another specific liver receptor

LCAT = Lecithin–cholesterol acyltransferase.
LPL = Lipoprotein lipase.

Table 67.2 The four main lipoproteins and their functions

Lipoprotein	Main apolipoproteins	Function
Chylomicrons	B_{48}, A-I, C-II, E	Largest lipoprotein. Synthesised by gut after a meal. Not present in normal fasting plasma. Main carrier of dietary triglyceride.
Very-low-density lipoprotein (VLDL)	B_{100}, C-II, E	Synthesised in the liver. Main carrier of endogenously-produced triglyceride
Low-density lipoprotein (LDL)	B_{100}	Generated from VLDL in the circulation. Main carrier of cholesterol
High-density lipoprotein (HDL)	A-I, A-II	Smallest lipoprotein. Protective function. Takes cholesterol from extrahepatic tissues to the liver for excretion (reverse cholesterol transport).

The endogenous lipid cycle

The liver synthesises VLDL particles that undergo the same form of delipidation as chylomicrons by the action of lipoprotein lipase. This results in the formation of an intermediate-density lipoprotein (IDL), which becomes low-density lipoprotein (LDL) when further delipidated. LDL may be removed from the circulation by the high-affinity LDL receptor, or by other scavenger routes that are important at high LDL levels and a key way in which cholesterol is incorporated into atheromatous plaques.

High-density lipoprotein (HDL) particles are derived from both liver and gut. They act as cholesteryl ester shuttles, removing the sterol from the peripheral tissues and returning it to the liver. The HDL is taken up either directly by the liver, or indirectly by being transferred to other circulating lipoproteins, which then return it to the liver. This process is thought to be anti-atherogenic, and an elevated HDL cholesterol level has been shown to confer a decreased risk of coronary heart disease on an individual.

Apolipoproteins

Apolipoproteins are the protein components of the lipoproteins (see Table 67.1). They are important in:

- maintaining the structural integrity of the lipoproteins
- regulating certain enzymes that act on lipoproteins
- receptor recognition.

The LDL receptor

The LDL receptor (Fig. 67.3), a glycoprotein present on the surface of all

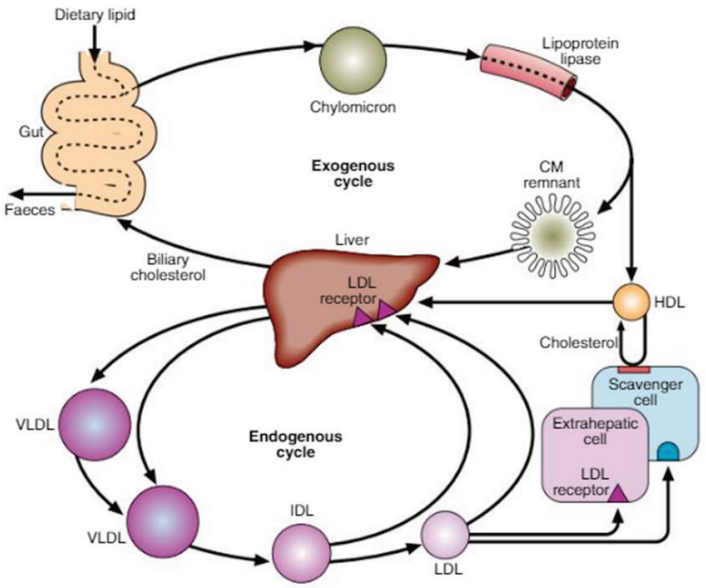


Fig. 67.2 Lipoprotein metabolism.

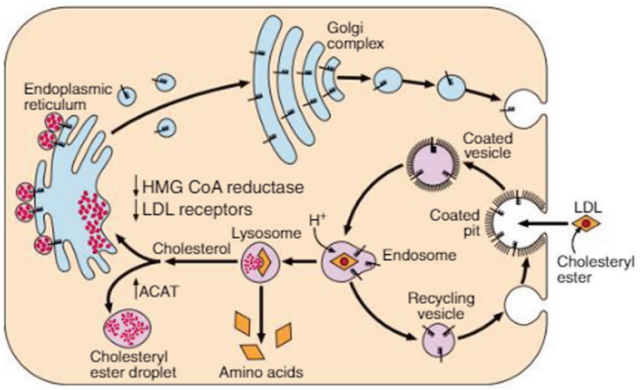


Fig. 67.3 The LDL receptor pathway. *ACAT* = Acyl-CoA:cholesterol acyltransferase; *HMG CoA* = hydroxymethyl glutaryl coenzyme A.

cells, spans the cell membrane and is concentrated in special membrane recesses, called 'coated' pits. It binds to lipoproteins containing apolipoprotein B and E, and internalises them for breakdown within the cell. Receptors are then recycled to the cell surface. The number and function of receptors dictate the level of circulating LDL. When the cell has sufficient cholesterol, the synthesis of receptors is down-regulated; when the cell is cholesterol depleted, the receptors increase in number. Inherited malfunction or absence of these receptors leads to familial hypercholesterolaemia (FH).

A specific mutation of apolipoprotein B results in defective binding of LDL to its receptor and produces an identical clinical picture to FH called familial defective apo B (FDB).

Clinical note

About 25% of the UK population have plasma cholesterol concentrations above the desirable level. In most cases this is the result of diet and lifestyle.

Lipoprotein metabolism

- Lipoproteins are complexes of lipid and proteins that facilitate lipid transport.
- Their metabolism can be thought of as two interconnected cycles centred on the liver.
- Lipoproteins are defined by their density and differ in composition, structure and function.
- Apolipoproteins have a functional as well as structural importance.
- Cholesterol can be excreted from the body only by way of the liver.

Case history 50

A 3-year-old boy with a history of chronic abdominal pain was admitted as an emergency. His blood was noted to be pink in the syringe, and the serum was milky.

Serum osmolality was measured as 282 mmol/kg and amylase 1780 U/L.

His triglyceride was reported to be greater than 50 mmol/L.

Na^+	K^+	Cl^-	HCO_3^-	Urea	Glucose
			mmol/L		
103	3.8	70	20	3.1	5.2

- Why is there a discrepancy between the calculated and measured osmolality?
- What are the likely causes of the hypertriglyceridaemia?

Comment on page 169.

Want to know more?

http://www.msdmanuals.com/en-gb/professional/endocrine-and-metabolic-disorders/lipid-disorders/overview-of-lipid-metabolism#v989655

There is no shortage of online material on this topic. This resource is accessible and does not go into unnecessary detail. However, it is sponsored by a pharmaceutical company (MSD), and the guidelines and recommendations reflect a US perspective.